Supporting Information for

Droplet-engineered appendix-derived gut organoids enable controlled growth, differentiation and fusion

Zhengyu Wu[1, 3], Rongzhou Chen [2,3], Wanlong Wang[1], Chuqian Ruan[1], Haotian Sun[1], Edmund Y. Lam[2], Shaohua Ma[1,4]

1. Institute of Biopharmaceutical and Health Engineering (iBHE), Tsinghua Shenzhen International Graduate School (SIGS), Tsinghua University, Shenzhen 518055, China
2. Department of Electrical and Computer Engineering, University of Hong Kong, Pokfulam, Hong Kong SAR, China
3. These authors contributed equally
4. Corresponding Email: ma.shaohua@sz.tsinghua.edu.cn

This PDF file contains 8 main items including:

Figures S1 to S7

Tables S1

And:

Legends for Source Data-1,2,3,4,5,6

SI References

Other supporting materials for this manuscript include the following:

Source Data-1

Source Data-2

Source Data-3

Source Data-4

Source Data-5

Source Data-6

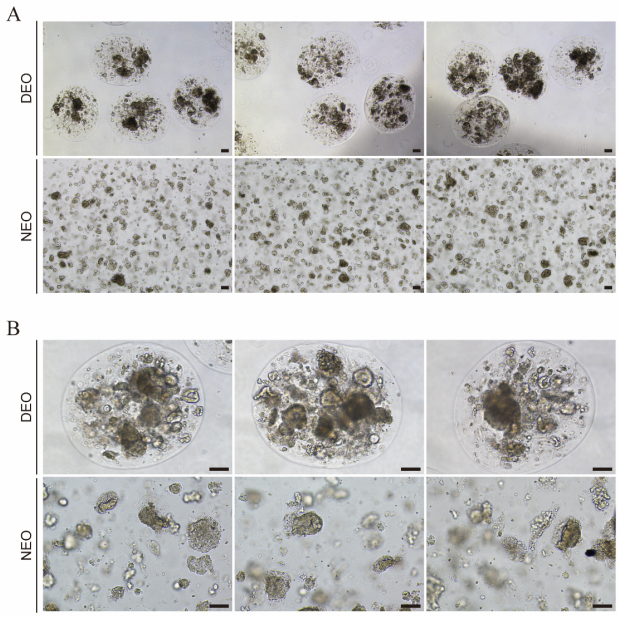


Figure S1 The comparisons between droplet-engineered organoid (DEO) and nonengineered organoid (NEO). Representative brightfield images of Day 0 seeding as crypts in Matrigel domes. Images were captured by using 4× objective lenses (A) and 10× objective lenses (B). Scale bars are 100 μm.

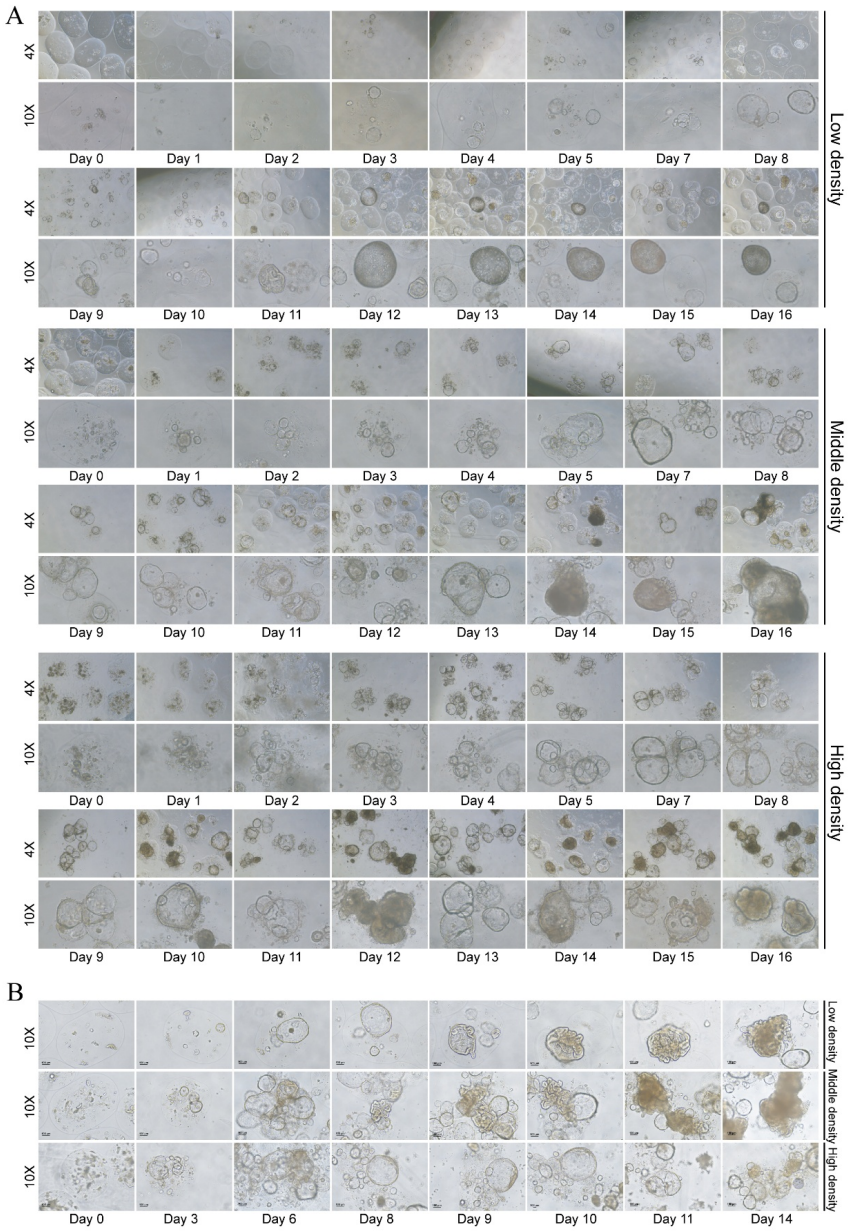


Figure S2 Continued growth of DEOs using low-density, middle-density, and high-density crypt suspensions. Representative images were captured from day 0 to day16. Images were captured by 4× and 10× objective lenses.

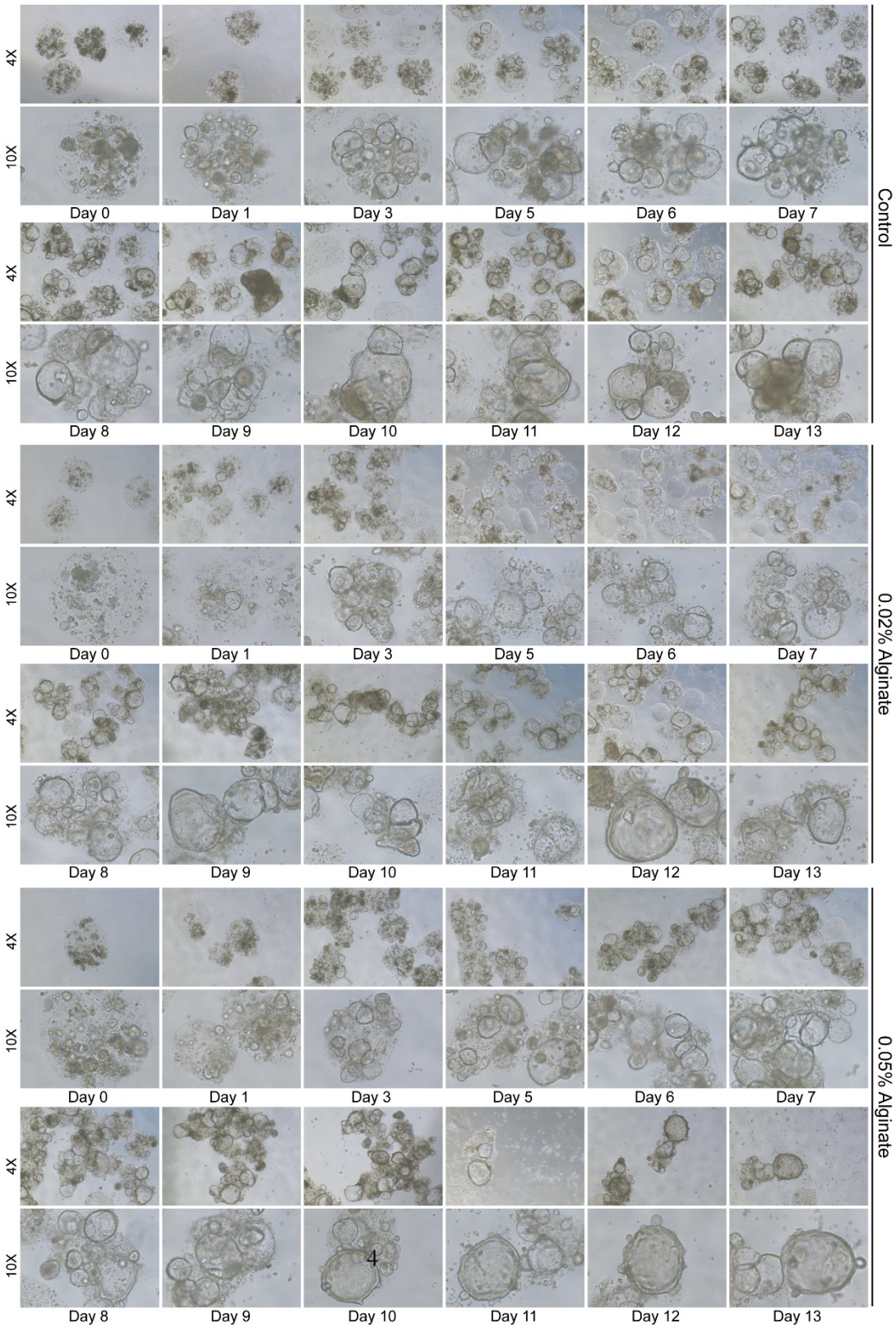

Control
0.02% Alginate
0.05% Alginate
4X
10X
Day 0
Day 1
Day 3
Day 5
Day 6
Day 7
Day 8
Day 9
Day 10
Day 11
Day 12
Day 13

Figure S3 Alginate incorporation of droplet-engineered organoids. DEOs in Matrigel alone (control) or Matrigel adding sodium alginate at final concentrations of 0.02% (low alginate) or 0.05% (high alginate). Representative images were captured from day 0 to day 13. Images were captured by 4× and 10× objective lenses.

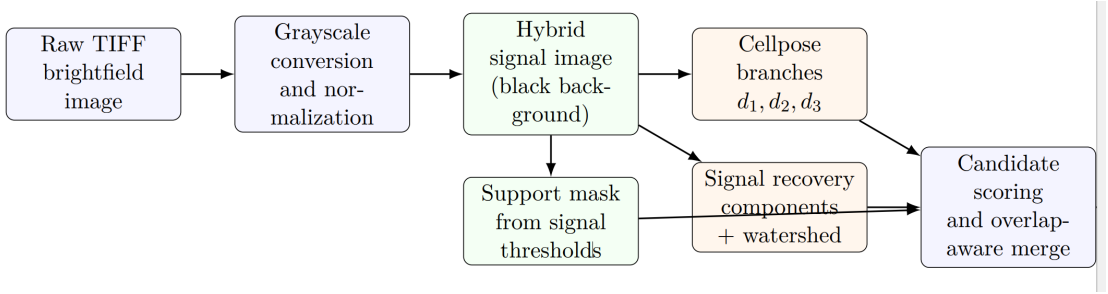


Figure S4 DEO segmentation workflow

The segmentation workflow generated final instance label maps from brightfield images using hybrid signal preprocessing, multiscale Cellpose branches, and signal-based large-mass recovery, followed by overlap-aware merging of candidate objects

（重新做一个好看的放正文）

（Figure S4-S7 放正文）

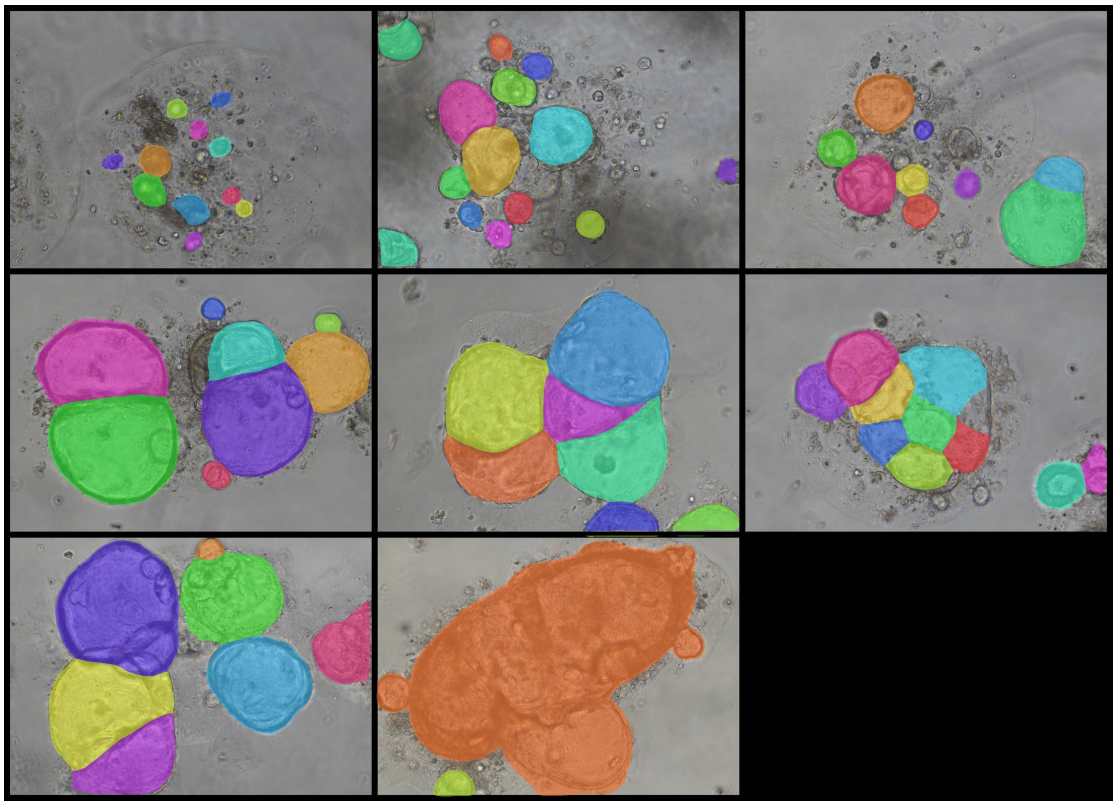


Figure S5 Representative segmentation outputs for density experiments

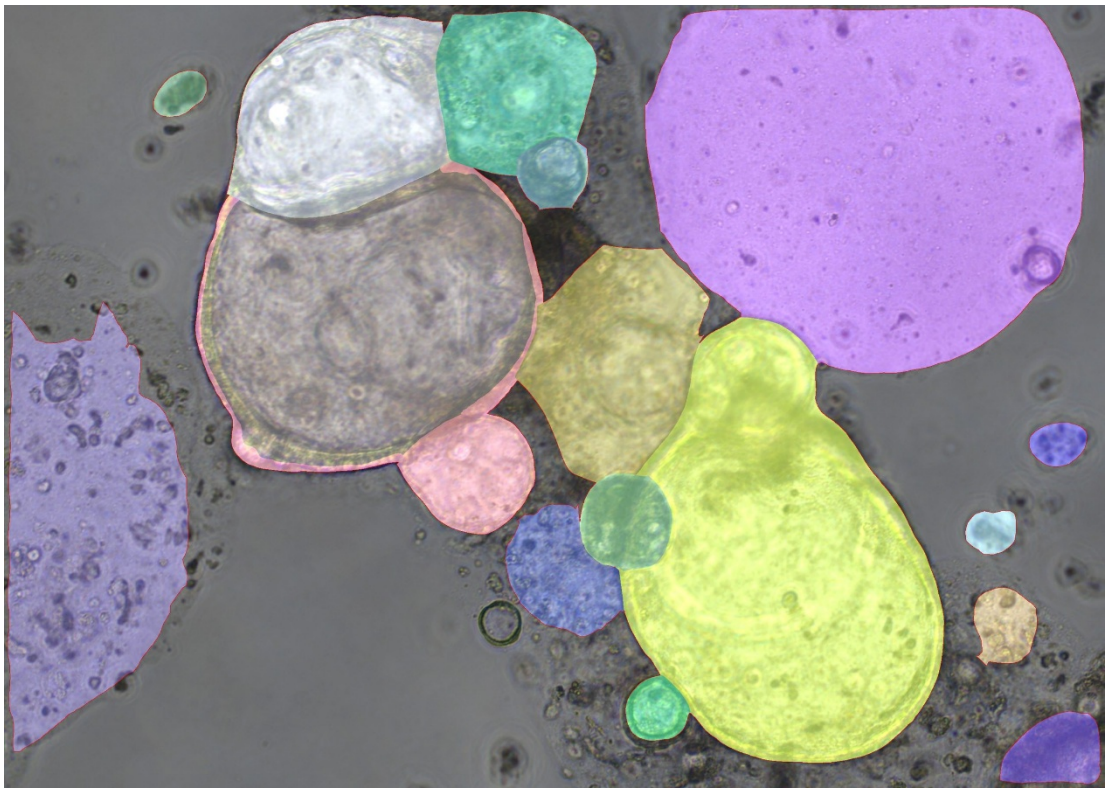


Figure S6 Representative segmentation outputs for alginate experiments

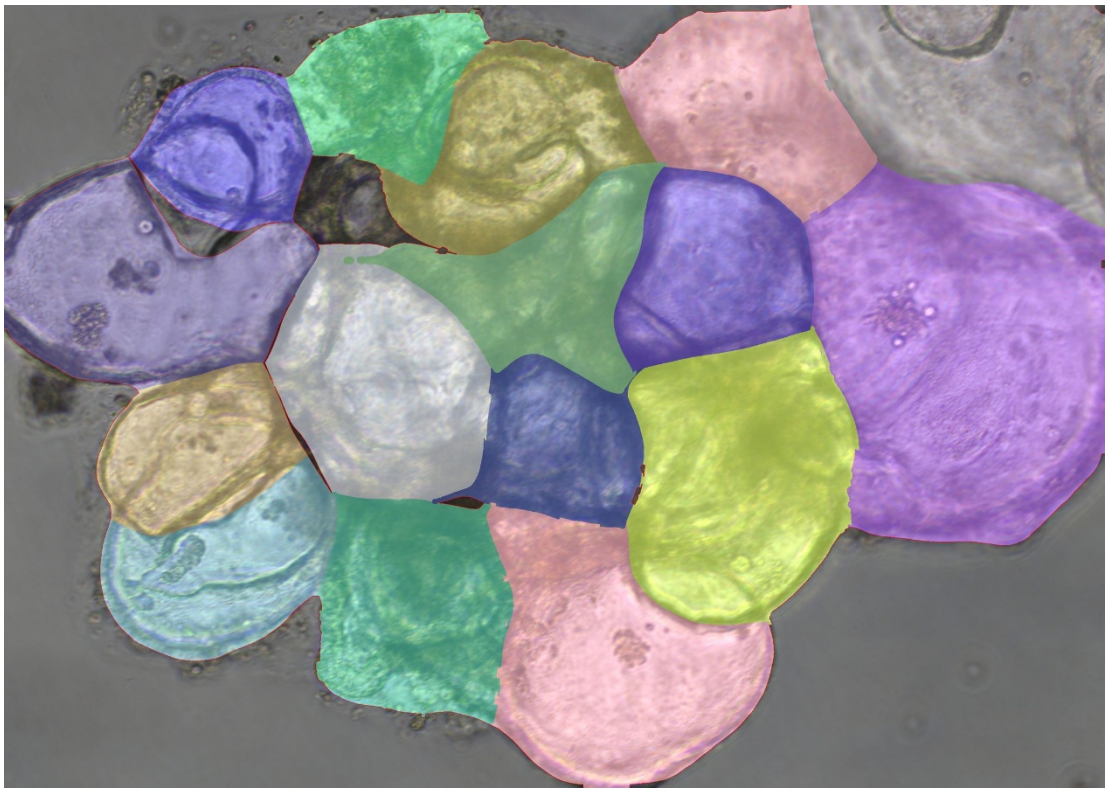


Figure S7 Representative segmentation outputs for Y-27632 experiments

Table S1

	Growth metric	Differentiation metric	Fusion metric	Rationale
Density	Total segmented area	Organoid darkness P90	Peripheral-to-central internal edge ratio	Total area is the main segmentation-based growth proxy. Darkness P90 captures darker, more differentiated morphologies. The peripheral-to-central ratio reflects redistribution of edge structure from the organoid periphery toward the interior during fusion.
Alginate	Average perimeter	Wall darkness P90	Peripheral-to-central internal edge ratio	Average perimeter provides a morphology-sensitive readout of organoid expansion. Wall darkness P90 captures darker, thickened epithelial walls associated with differentiation. The same fusion metric was used because it reports the relative shift of edge signal from peripheral regions to the interior.

	Growth metric	Differentiation metric	Fusion metric	Rationale
Y-27632	Mean organoid area (derived from segmented instance areas)	Organoid darkness P90	Center-weighted edge sum	Mean organoid area summarizes organoid size while reducing confounding from fusion-related changes in object number. Darkness P90 captures differentiation-associated darkening. The center-weighted edge sum emphasizes interior edge structure and is therefore sensitive to fusion-associated structural integration.

Abbreviations: P90, 90th percentile; DEO, droplet-engineered organoid.

Source Data 1. Statistical analysis of density experiments at matched time points

Source Data 2. Pairwise statistical comparisons between density groups and control at matched time points

Source Data 3. Statistical analysis of alginate experiments at matched time points

Source Data 4. Pairwise statistical comparisons between alginate groups and control at matched time points

Source Data 5. Statistical analysis of Y-27632 experiments at matched time points

Source Data 6. Pairwise statistical comparisons between Y-27632 groups and control at matched time points

SI References

[1]

[2]